2025 WSMO Tiebreaker Round Solutions

SMO Team

Tiebreaker Round Problem 1: Find the sum of the digits of $2499^2 + 2501^2$.

Answer: $\boxed{10}$

Solution: Note that

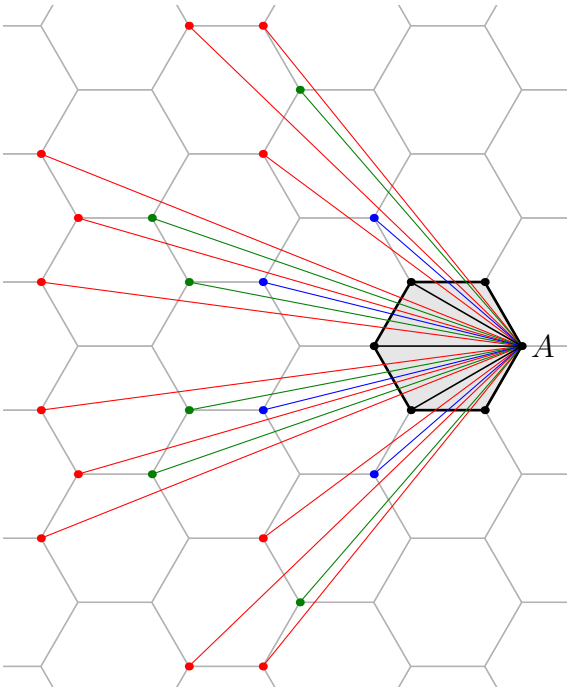
$$2499^2 + 2501^2 = (2500-1)^2 + (2500+1)^2 = 2 \cdot 2500^2 + 2 = 12500002,$$

so the sum of the digits is $1+2+5+2 = \boxed{10}$.

Tiebreaker Round Problem 2: In a regular hexagon, a beam of light is shot from one vertex of the hexagon. It bounces around inside the hexagon until it hits a vertex of the hexagon. Including the two directions along the sides of the hexagon, how many angles can the light be shot at such that it bounces at most three times before hitting a vertex?

Answer: $\boxed{27}$

Solution:



Let the hexagon have side length 1, and let A be the vertex the light is shot from. We reflect the hexagon across its sides repeatedly, which tiles the plane with regular hexagons. Each path of light then unfolds into a straight segment from A to a vertex V of the tiling that

passes through no other vertex, and the number of bounces is the number of sides of the tiling that the segment crosses. Place A at the origin with the hexagon to its left. Checking the vertices of the tiling near A (shown above), the possible endpoints V are

$$\left(-\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right), \left(-\frac{3}{2}, \pm\frac{\sqrt{3}}{2}\right), (-2, 0) \qquad \text{for 0 bounces,}$$

$$\left(-2, \pm\sqrt{3}\right), \left(-\frac{7}{2}, \pm\frac{\sqrt{3}}{2}\right) \qquad \text{for 1 bounce,}$$

$$\left(-3, \pm 2\sqrt{3}\right), \left(-5, \pm\sqrt{3}\right), \left(-\frac{9}{2}, \pm\frac{\sqrt{3}}{2}\right) \qquad \text{for 2 bounces, and}$$

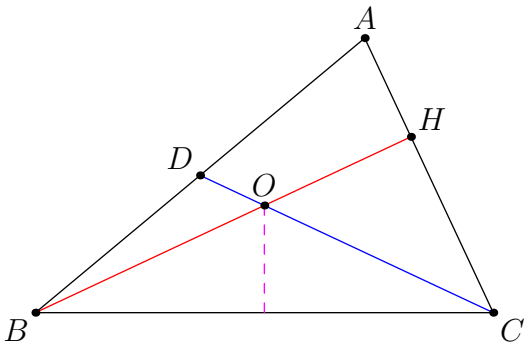
$$\left(-\frac{7}{2}, \pm\frac{5\sqrt{3}}{2}\right), \left(-\frac{9}{2}, \pm\frac{5\sqrt{3}}{2}\right), \left(-\frac{7}{2}, \pm\frac{3\sqrt{3}}{2}\right),$$
$$\left(-\frac{13}{2}, \pm\frac{3\sqrt{3}}{2}\right), \left(-6, \pm\sqrt{3}\right), \left(-\frac{13}{2}, \pm\frac{\sqrt{3}}{2}\right) \qquad \text{for 3 bounces.}$$

Note that the 5 directions with no bounces include shooting the light along the two sides of the hexagon at A. So, the answer is $5 + 4 + 6 + 12 = \boxed{27}$.

Tiebreaker Round Problem 3: In $\triangle ABC$, D is the midpoint of $\overline{AB}$. The perpendicular bisector of $\overline{BC}$, the altitude from B to $\overline{AC}$, and line CD are concurrent. If $\angle BAC = 3\angle DCB$, the value of $\cos(\angle DCB)$ can be expressed as $\frac{\sqrt{m}+\sqrt{n}}{p}$, where m, n, and p are positive integers and m and n are not divisible by the square of any prime. Find $m + n + p$.

Answer: $\boxed{44}$

Solution:



Let O be the point of concurrency and let $\theta = \angle DCB$. Since O lies on the perpendicular bisector of BC, we have $\angle OBC = \angle OCB = \theta$. Since $BO \perp AC$, we have $\angle ACB = 90^\circ - \theta$. So, $\angle BAC = 3\theta$ and

$$\angle ABC = 180^\circ - 3\theta - (90^\circ - \theta) = 90^\circ - 2\theta,$$

meaning $\angle ACD = \angle ACB - \theta = 90^\circ - 2\theta$. Since D is the midpoint of AB, we have $[ACD] = [BCD]$, so from the sine area formula,

$$AC \cdot \sin\angle ACD = BC \cdot \sin\angle BCD \implies \frac{AC}{BC} = \frac{\sin\theta}{\cos 2\theta}.$$

From the Law of Sines on triangle ABC, we also have

$$\frac{AC}{BC} = \frac{\sin \angle ABC}{\sin \angle BAC} = \frac{\cos 2\theta}{\sin 3\theta}.$$

So,

$$\cos^2 2\theta = \sin\theta \sin 3\theta = \frac{\cos 2\theta - \cos 4\theta}{2} = \frac{\cos 2\theta - 2\cos^2 2\theta + 1}{2} \implies$$
$$0 = 4\cos^2 2\theta - \cos 2\theta - 1.$$

Since $\angle ABC = 90^\circ - 2\theta > 0$, we have $\cos 2\theta > 0$, so $\cos 2\theta = \frac{1+\sqrt{17}}{8}$. Thus,

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \frac{\sqrt{9+\sqrt{17}}}{4} = \frac{\sqrt{2}+\sqrt{34}}{8} \implies 2+34+8 = \boxed{44}.$$